

Hemant Thapa

AI/ ML Full stack Engineer

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SUMMARY

Senior Machine Learning & AI Engineer with +5 years of hands-on experience building and implementing machine learning and AI-based solutions for large enterprises, including developing LLMs, RAG pipeline, generative AI and other related solutions. I have built an end-to-end MLOps system to support AI/ML solutions for achieving production-level results and impacting business in a positive manner.

SKILLS

Artificial Intelligence & Machine Learning

Supervised & Unsupervised Learning, Model Evaluation Metrics, Feature Engineering, Deep Learning, XGBoost, A/B Testing, Causal Inference, Generative AI Workflows

Large Language Models & Generative AI

Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Amazon Agents, Prompt Engineering, Natural Language Processing (NLP), Syntactic Analysis, Sampling Methods

AI Safety, Guardrails & Evaluation

Prompt safety, hallucination detection, retrieval grounding, offline/online evaluation

ML Libraries & Frameworks

PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Apache Spark, Matplot, Seaborn

MLOps & Model Deployment

MLOps Pipelines, Model Training, Versioning & Deployment, CI/CD for ML, Kubeflow, Containerized Inference Services

Cloud ML Platforms

AWS SageMaker, GCP Vertex AI, Azure Machine Learning

Cloud & Infrastructure

AWS, GCP, Azure, Docker, Kubernetes, Terraform, Pulumi, CloudFormation

Monitoring & Observability

CloudWatch, Grafana, New Relic

Data Engineering (ML Support)

Spark, Kafka, Flink, Airflow, SQL, Databricks

Programming Languages & Frameworks

Python, Go, JavaScript/Typescript, Node.js, React.js, React Router, Vite, Express.js, REST APIs, Java, C#, .NET, C++, Node, API Orchestration, Postman

Data Query Language (DQL)

SQL, PostgreSQL, MySQL, MariaDB, SQLite, Spark, MongoDB, Firebase, Prisma

PROFESSIONAL EXPERIENCE

AI / ML Engineer

Costrick

04/2024 – Present

Glasgow, Scotland

Led development of AI-powered decisioning and agent-like systems for real-time credit and risk workflows using Python, Java, and Go, integrating ML models and LLM components into low-latency microservices exposed via REST/gRPC APIs. Designed and iterated on intelligent workflows combining predictive models and LLM-based reasoning, leveraging structured prompting, tool usage patterns, and retrieval-augmented generation (RAG) to enable context-aware decisioning and internal automation.

Built and enhanced predictive models using feature engineering on behavioral and transactional data (pandas, NumPy, SQL), with production-grade offline/online validation pipelines and low-latency inference services in Java/Go.

Developed and productionized LLM-based internal tools using OpenAI APIs, LangChain/LlamaIndex, vector databases (FAISS, Pinecone), and modular pipelines approximating multi-step agent behavior with external tool integration.

Implemented scalable backend systems (Spring Boot, Node-compatible APIs) supporting high-throughput decisioning, agent-to-service interactions, and integration with internal risk and compliance platforms.

Strengthened fraud detection and anomaly monitoring using supervised and unsupervised learning, real-time inference, and stream processing (Kafka/Kinesis), enabling responsive, event-driven decision systems.

Established observability for ML and AI systems, including tracing, logging, cost monitoring, drift detection, and failure analysis using Prometheus, CloudWatch, and custom metrics to ensure reliability of production AI workloads.

Designed experimentation and evaluation frameworks, including A/B testing and offline validation, to measure model and AI system performance and support iterative improvements.

Collaborated cross-functionally with product, risk, and compliance teams to ship compliant, explainable AI systems (SHAP/LIME) with strong auditability and governance controls.

Leveraged AI-assisted development tools (Claude, Cursor, Windsurf) as primary coding accelerators for rapid prototyping, code generation, and refactoring across distributed microservices.

Founder & AI Engineer

Squid Consultancy Group (SCG)

08/2023 – Present
Glasgow, Scotland

Designed and optimized RAG-based GenAI systems using vector databases (Milvus, Pinecone, Weaviate) and advanced retrieval techniques (chunking, hybrid search, reranking) to power context-aware, real-time recommendations and intelligent design assistant workflows.

Built LangChain-based RAG and agent pipelines (retrievers, tools, prompt templates, chains) enabling knowledge-grounded generation, structured outputs, and tool-augmented workflows for collaborative design and content generation use cases.

Architected agentic workflows using LangGraph with graph-based orchestration, supporting multi-step reasoning, tool calling, retries/fallbacks, state management, and memory for persistent, interactive user-facing agents.

Developed and productionized agent-driven GenAI systems with modular components for reasoning, tool integration, and memory, enabling autonomous task execution and adaptive user support.

Implemented scalable multi-agent-like workflows by composing specialized components (retrievers, planners, executors) with clear interfaces, approximating agent orchestration patterns and enabling extensibility for new tools and capabilities.

Led large-scale data ingestion and streaming pipelines using Apache Spark, Flink, Kafka, and Airflow, supporting continuous data flow for training, evaluation, and real-time inference on design and collaboration datasets.

Deployed production-grade MLOps and LLM pipelines using Kubeflow, CI/CD, Docker, and Kubernetes, enabling versioned releases, automated retraining, and scalable inference across distributed systems.

Developed microservices exposing ML and LLM capabilities using Python, Java (Spring Boot), C#, and Go, with secure, scalable APIs deployed on ECS/EKS, Azure Kubernetes Service, and GCP GKE.

Built high-throughput backend systems for model serving orchestration and agent-to-tool interactions using Java (Spring Boot), gRPC, and distributed service architectures.

Implemented AI system observability and reliability practices, including tracing, structured logging, cost monitoring, latency tracking, and failure detection using CloudWatch, Grafana, and New Relic.

Established evaluation and testing workflows for GenAI systems, including offline benchmarking, prompt/response validation, and iterative experimentation to improve agent reliability and output quality.

Applied robust security and governance controls for LLM systems, including prompt-injection mitigation, tool access controls (allowlists), least-privilege IAM, input/output sanitization, jailbreak resistance, and PII detection/redaction.

Integrated multiple LLM providers (OpenAI, Google Gemini, Anthropic Claude) into production systems, enabling flexible model routing, experimentation, and performance optimization.
Leveraged AI-assisted development tools (Claude, Cursor) within CI/CD workflows for code generation, automated reviews, testing, and rapid iteration on agent-based systems.
Partnered with product, design, and engineering teams to define agent workflows, tool interfaces, data contracts, and platform intelligence requirements for production deployment.

Future Analyst

Demola Global

03/2023 – 05/2023
Paisley, Scotland

Extensive research was conducted to predict the evolving impact of artificial intelligence across industries and society. A comprehensive machine learning library was developed from scratch, capable of handling tasks such as linear and non-linear regression, classification, PCA, exponential moving averages, k-means clustering, and feature scaling.

As part of the Demola Group project, a diverse, multicultural team was managed to build inclusive AI solutions while balancing full-time programming and volunteer research responsibilities. The team created a user-friendly tool combining a linear regression model with a scatter plot, integrated with a voice assistant, enabling non-technical users to easily interpret market trends and stock prices. Success was achieved through regular collaboration, effective time management, and a focus on accessibility.

Contributions were recognized with European Credit Transfer and Accumulation System (ECTS) credits from Kajaani University of Applied Sciences (KAMK) for work as a Future Analyst. This experience further strengthened expertise in AI, machine learning, and market analysis while fostering meaningful contributions to the global community.

Full Stack Engineer

Custom Patches & Wear Ltd

08/2021 – 02/2023
Glasgow, Scotland

Championed the transition of our JavaScript applications from Backbone.js to React, enabling orders of magnitude greater team productivity and application reliability
Integrated lightweight ML models (e.g., XGBoost) into React apps for real-time personalization and recommendation features

Helped increase the number of contributors to our JavaScript applications to include ~90% of our engineering team

Worked across disciplines to onboard engineers of all skill levels into on-call rotations

Architected and built applications across a wide number of business units, from customer-facing applications to internal CRMs and productivity tools, incorporating data pipelines with Kafka for ML-driven insights

Helped spearhead our engineering internship program and campus recruiting endeavors

EDUCATION

Online Data Science Bootcamp

HyperionDev

The Department for Education (DfE) Fully Funded Data Science

11/2022 – 02/2023
Cape Town, South Africa

Bachelor of Engineering (Hons)

University of the West of Scotland

08/2017 – 07/2021
Paisley, Scotland

PROJECTS

nalyst 2.1.2

Full-stack machine learning, statistics, time series, survival, and deep learning toolkit with a unified train/infer API, AutoML, explain ability, and diagnostics.

03/2023 – Present

<https://pypi.org/project/nalyst/> 

scg-lab 5.5.7 <i>SCG Lab is a browser-based notebook environment that runs entirely on your machine. It bundles a React/Monaco frontend with a FastAPI + Jupyter-kernel backend into one installable Python package.</i> pypi.org/project/scg-lab/5.5.7	05/2024 – Present
AI Job Tracking <i>AI Job Tracking is part of our suite of automation tools designed to help professionals save time and get ahead. Our team of developers, designers, and strategists build products that make a real difference.</i> www.aijobtracking.com	06/2024 – Present
kriyatha <i>Kriyatha is a SaaS platform, which brings planning, tracking, collaboration, and billing into one workspace your whole team will actually use.</i> www.kriyatha.com	08/2024 – Present
neurova.js <i>Build full-stack TypeScript products from one cohesive ecosystem instead of stitching together disconnected tools. A unified UI, Backend, AI framework MERN stack build from scratch for modern web apps.</i> https://www.npmjs.com/search?q=neurova	09/2024 – Present
algorithm-stat 0.1.2 <i>A self-contained Python toolkit for market data, metrics, charts, Monte Carlo, backtesting, ML, macro, options, mortgage, and Alpaca paper trading.</i> https://pypi.org/project/algorithm-stat	11/2024 – Present
scg bot services <i>SCG Bot Services creates intelligent AI chatbots that learn from your website automatically. Answer visitor questions, provide instant support, and boost engagement</i> www.scgbotservices.com	10/2025 – Present
Neurova <i>Python package for image processing, computer vision, deep learning, and classical machine learning. Complete computer vision and deep learning toolkit with GPU acceleration.</i> pypi.org/project/neurova/	02/2026 – Present
scg-auth 1.1.1 <i>A lightweight, zero-dependency OAuth 2.0 client library, Authorization Code, Client Credentials, Implicit, and Device Code flows with PKCE and CSRF protection.</i> www.npmjs.com/package/scg-auth pypi.org/project/scg-auth/1.1.1	03/2026 – Present
crypto-gate 1.1.1 <i>Consistent API covering password hashing, symmetric/asymmetric encryption, digital signatures, hashing, key derivation, and secure token generation.</i> pypi.org/project/crypto-gate/ www.npmjs.com/package/crypto-gate	05/2026 – Present
scg-jwt-token 2.0.1 <i>A simple, secure, and universal JWT library for web and mobile applications.</i> www.npmjs.com/package/jwt-scg pypi.org/project/scg-jwt-token/2.0.1/	05/2026 – Present

CERTIFICATES

Google Data Analytics

IBM Data Science

Future Analyst

Mathematics for Machine
Learning and Data Science